

## Homework 2

Release Date: 1404/8/11

Deadline: 1404/8/17

### 1 Hilbert Transform (25 pts)

Find the Hilbert transform of following signals.

1.  $x_1(t) = \text{sinc}(2Wt)$ .
2.  $x_2(t) = A \cos(2\pi f_0 t + \theta)$ .
3.  $x_3(t) = A\Pi(\frac{t}{T}) = A[u(t + \frac{T}{2}) - u(t - \frac{T}{2})]$ .  
(a) show that if  $v(t) = A$  for all time, then  $\hat{v}(t) = 0$ .
4.  $x_4(t) = \frac{1}{(t-a)^n}$ .
5.  $x_5(t) = \cos^4(\omega t)$ .
6.  $x_6(t) = \frac{t}{(a^2+t^2)^2}$ .

### 2 Hilbert Transform Properties (15 pts)

1. Show that the Hilbert transform of an even signal is odd and the Hilbert transform of an odd signal is even.
2. Let  $x(t)$  represent a bandpass signal and  $m(t)$  denote a lowpass signal with nonoverlapping spectra. Show that the Hilbert transform of  $c(t) = m(t)x(t)$  is equal to  $\hat{c}(t) = m(t)\hat{x}(t)$ .

### 3 Even-Odd Decomposition (20 pts)

Suppose an LTI system with impulse response  $h(t)$  and transfer function  $H(f)$ . We assume that this filter is physically realizable, then its impulse response is real and has the causal property, i.e.,  $\forall t, h(t) \in \mathbb{R}$  and  $h(t) = 0$  for  $t < 0$ .

1. We define even part and odd part of  $h(t)$  as  $h_e(t) = \frac{h(t)+h(-t)}{2}$  and  $h_o(t) = \frac{h(t)-h(-t)}{2}$ , respectively.

Show that in this case, we can write

$$h(t) = 2h_e(t)u(t).$$

(Assume that  $u(0) = \frac{1}{2}$ .)

2. Write  $h_e(t)$  in terms of  $H_e(f)$ , the Fourier Transform of  $h_e(t)$ .
3. Write  $h(t)$  in terms of  $H_e(f)$ .
4. Write  $h_o(t)$  in terms of  $H_e(f)$ .
5. Write  $H_o(f)$  in terms of  $H_e(f)$ .
6. Show that  $H_o(f) = -j\hat{H}_e(f)$ , i.e.,  $H_o(f) = \left(\frac{1}{j\pi f}\right) * H_e(f)$ .

#### 4 Wireless Connection (20 pts)

A network with one Base Station (BS) and two users is shown in Fig. 1. The distance between user 1 and the BS is 1 km, and the distance between user 2 and the BS is 2 km. The transmitted power of user 1 and user 2 are 20 dBm and 36 dBm, respectively. The free-space path loss is modeled as

$$L_{FS}(dB) = 20 \log(d) + 20 \log(f) - 147.55$$

, where  $d$  is the distance between the BS and the user in meters, and  $f$  is the frequency in Hz.

- a) For which of the users can a connection to the BS be established? Assume  $f = 2.1$  GHz and the minimum received power at the BS that can be detected is  $-70$  dBm.
- b) Now, assume that the user who is unable to establish a connection with the BS employs an amplifier with gain  $\alpha$  at its location. What should be the value of  $\alpha$  so that the BS can successfully detect the user's signal?
- c) Instead of using an amplifier, suppose that this user attempts to establish a connection at a different frequency. Determine the frequency required to establish the connection and discuss whether this is a practical solution.

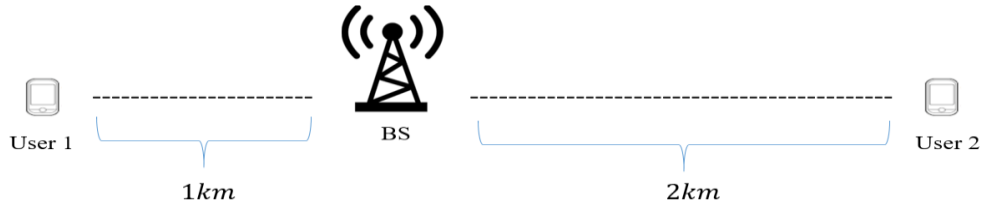


Figure 1: A network with the BS and users.

#### 5 Attenuation and Amplification (20 pts)

Let  $x(t)$  be a periodic signal depicted in Fig. 2.

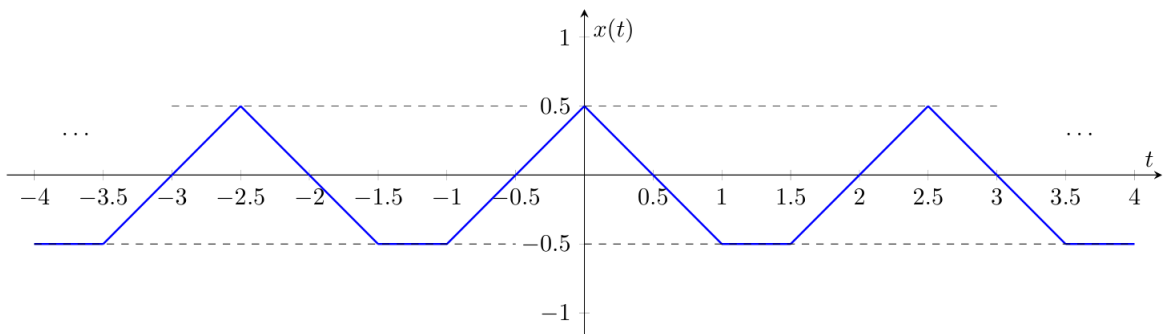


Figure 2: A periodic signal

1. Find  $X(f)$ , the Fourier transform of  $x(t)$ .
2. Find  $P_x$ , the power of  $x(t)$ .
3. Under the assumption that  $P_x$  is in *watts* suppose the signal  $x(t)$  is to be transmitted via a 500 km repeater system consisting of  $m$  identical fiber optic cable sections with attenuation  $\alpha = 2$  dB/km and  $m$  identical amplifiers with gain  $\beta$ . Find the *minimum* required number of sections and gain per amplifier so that  $P_{out} = P_{in}$ , and the input power to each amplifier is at least  $20 \mu\text{W}$ .

## Submission

Please submit your answers in a .pdf file named CS\_HW2\_studentID.pdf, replacing studentID with your actual student ID.